

SOLAR SYSTEMS WIRING

Joints and Terminations

Cable preparation

IEE regulations requires that electrical joints must be mechanically and electrically sound. When making electrical joints, the first step in preparing the end of a cable for entry into an accessory, such as a switch, a lamp holder, or a ceiling rose, is to remove insulation for a short length of the conductor. In addition, if it is rubber insulated cable, the braid and tape must be stripped back for approximately 15 mm from the insulation. This is because the braid and tape are intended for mechanical protection, not for insulation, and in a damp situation may become conductors and cause leakage from the conductor. In the case of rubber insulated cables, the stripping of braid, tape and insulation is generally done with a sharp blade or electrician's knife. The blade should be held at an acute angle to the cable so that the insulation is pared rather than cut. To cut at a near right angle would be very bad practice because, although it may give a neater finish, it would probably cause nicking of insulation and conductor. A nicked conductor becomes so weak that, after being bent a few times, it will almost certainly break. Apart from the tendency to break, the effective cross-sectional area of the conductor will be reduced, causing increased resistance which may result in excessive heat and eventually fire.

Cable Termination

- Termination is the entry of a cable end into an accessory. In case of a stranded conductor, the strands should be twisted tightly together before termination. Great care must be taken not to damage the wires. The I.E.E. Regulation requires that a cable termination of any kind shall securely anchor all the wires of the conductor and shall not impose any mechanical stress on the terminal or socket. If a soldering socket is used, then it must contain all the strands of the conductor.
- A termination under mechanical stress is liable to disconnection. When current is flowing in a cable, certain amount of heat is developed, and the consequent expansion and contraction may force the cable out of the terminal or socket.
- If one or more strands of the cable were to be left out of the terminal or socket, there would be a reduction in the effective cross-sectional area of the conductor at the point. When terminating rubber insulated cable, the technician should avoid exposing the rubber insulation and conductors to the atmosphere. In conditions of heat or moisture, the terminations should be sealed.
- Mineral insulated copper sheathed cables are terminated in *sealed glands*. A common arrangement is the screw-on pot-type seal, in which a small metal pot is screwed on to the copper sheath. Cold plastic compound is then pressed into the pot and a fiber cap and insulating sleeves are threaded over conductors and crimped into position.
- If an insulation test indicates that moisture has entered the mineral insulation, the gland must be undone, the moisture driven out using a blowlamp and the seal re-made.

Types of terminals

- The most commonly used methods of securing conductors in accessories are:

Pillar terminals,

- A pillar terminal has a hole through its side into which the conductor is inserted and then secured by a set screw. If the conductor is small in relation to the hole, it should be doubled back. When two or more conductors are to go into the same terminal, they should first be tightly twisted together. In the case of flexible cord terminations, the strands must be twisted up (and then bent back if room permits) before being entered into the terminal.

Screw Head, Nuts and Washers.

- When fastening conductors under screw heads or nuts, it is best to form the conductor end into an eye by means of the round-nosed pliers. The eye should be slightly larger than the screw shank, but smaller than the outside diameter of the screw head, nuts or washers. The eye should be placed in such a way that rotation of the screw head or nut tends to close the joint in the eye. If the eye is put the opposite way round, the motion of the screw or nut will tend to untwist the eye, and will probably result in imperfect contact.
- The importance of good contact cannot be over-stressed. Poor contact introduces resistances at the termination and leads to over-heating eventually leading to fire.
- *Solder less lug connections* are now being used to some extent in the electrical contracting industry, particularly for terminating small sized cables. Lugs of this type are made from solid copper and are tinned. They are fastened to cable ends by crimping. Hand crimping tools usually accommodate several cable sizes in a fixed head.
- For the larger sized lugs, there are hydraulic crimping machines with capacities of several tonnes.

Termination Tools

- The following tools are required while terminating an electrical cable
- A sharp knife,
- A pair of side-cutters and
- A soldering iron.
- The knife should be sharpened periodically. The cutting edge of pliers and side-cutters should not be used for purposes other than that for which they are intended. The bit of the soldering iron should be kept clean.
- For compliance with the I.E.E. Regulations, joints in cables must be;
- Made by soldering or by mechanical clamps
- Mechanically and electrically sound
- Provided with insulation not less effective than that of the cable cores,
- Protected against moisture and mechanical damage
- Accessible for inspection (except in the case of buried cables)

Soldering

- The type of solder to be used for twist-joints is called 'tin man's solder.' It is a soft solder consisting of 60 parts tin to 40 parts leads, with a melting point of 200°C (about one-fifth of the melting point of copper). To make the solder flow readily over a surface, a *flux* is required. The function performed by the flux is largely that of cleaning the surface and of reducing any oxide to the metallic state.
- For electrical work, pure amber resin is the safest flux to use, because it does not cause any corrosion of the metal. Soldering fluxes containing acid or other corrosive substances, such as zinc chloride solution (killed spirits) must not be used for this work

The Soldering iron

- When the soldering iron is heated, the surface should be cleaned, a small quantity of resin is applied and then the stick of solder lightly touched. The surface should then have a bright silvery appearance. Powdered resin is applied to the joint which is then laid in a shallow pool of solder. After a short period, the solder will run up into the strands, at which point joint can be removed from the iron and allowed to cool. Sometimes, soldering bits are grooved so that the portion of joint to be soldered can be placed in the groove, thus giving a larger area of surface contact between the two.
- A well-soldered joint should present a smooth, bright appearance. It is important to remember that wiping the join before it cools destroys this appearance and is liable to roughen the surface and become detrimental to the insulation.
- To avoid damaging the cable insulation near the joint the soldering process should be carried out as quickly as possible. The tendency to burn the insulation is less with a really hot soldering iron than with a cooler one which takes longer to do the job. It is sometimes possible to roll back some of the insulation before soldering and to unroll it afterwards.
- *The metal pot and ladle.* An iron metal pot about three quarters full of solder, together with the ladle, is heated until the solder is fluid. Any impurities can then be skimmed off the surface. The joint is held over the pot and sprinkled with resin a small amount being allowed to fall into the solder. The ladle should be filled with solder and this should be poured heavily over the joint so as to raise its temperature as rapidly as possible, thereby avoiding damage to the insulation. Pouring is repeated, adding resin as necessary, until the required coating is achieved; at the same time any points that may form are carefully brushed off.
- In the case of a termination, both the lug and the cable end must be heated and 'tinned' before any attempt is made to insert cable end in position.
- *The blowlamp.* Before starting to solder, see that the blow lamp is about two-thirds full of paraffin. For pouring, it is best to fit a funnel in the filler cap. A piece of cotton waste is then placed in the annular trough, soaked in methylated spirit, and ignited. It is possible to use paraffin for this purpose, but methylated spirit gives a cleaner flame and starts the lamp quicker. By operating the pump, the pressure inside the reservoir can be increased. This pump reaches nearly to the bottom of the reservoir and is provided with a non-return valve. As the pressure increases, paraffin will be forced up the tube and becomes vaporized by the heat from the

burning waste. Vapour, so formed, will then issue from the orifice (or jet) at high velocity and after mixing with air drawn in through holes at the side of the chamber, will burn at the nozzle.

- To enable the blowlamp to work continuously after it has once started, the tube conveying paraffin from the reservoir to the jet is led near the flame, so that the paraffin is vaporized by the heat of the lamp itself. The pressure inside the reservoir is also maintained by the heat of the lamp. As this pressure is high, the base of the lamp is curved inward to give additional strength.
- To stop the blowlamp working, the valve is opened. The flow of paraffin then stops due to the reduction in pressure. When using the blowlamp to solder a joint, the flame should be directed into the middle of the joint. As the conductor attains the melting temperature of solder, resin and solder, are rubbed into it alternately until it becomes saturated with solder.
- Because of the risk of causing damage, it is advisable to wrap the ends of the insulation with dampened cloth tape before soldering joints with the aid of a blowlamp. Before sweating a cable socket using a blowlamp the interior of the socket must be cleaned. The inside is then tinned by heating and introducing solder and resin. After almost filling the socket with molten solder, the cable end, rubbed with resin, is pushed into it. The solder should run up into the strands of the cable, and should be evenly dispersed round the conductor. The blowlamp jet should always be kept clear by using the pricker as necessary. It is bad practice to operate the pump to clean the jet, if the flame has gone out.
- Blowpipes utilizing propane or butane gases are gradually superseding paraffin blowlamps. Special soldering techniques are used for aluminium.

MAINTENANCE AND SERVICING OF SOLAR PV SYSTEM

Purpose of Maintenance

Maintenance is carried out on a solar PV system to ;

- i. Ensure that the system is always in good working condition.
- ii. Prevent faults and fix problems

The parts that need close maintenance are the battery and the solar PV module.

The approach of preventive maintenance is important and entails the following;

- i. Checking each component to see whether it is in good working condition or not.
- ii. Replacing faulty components.
- iii. Preventing future breakdown.
- iv. Replacing consumables

Battery Maintenance

The following maintenance procedure should be carried out on the battery.

Cleaning

This should be done once a month so that the battery is maintained clean. The following procedure is followed when doing the cleaning:

- i. Disconnect the battery from the charge controller
- ii. Remove the battery terminals from the posts
- iii. Clean the top and outside of the battery with clean water. Ensure that the water does not enter the battery cells
- iv. Clean the terminals and posts. If the terminals are corroded with white powder, use baking powder to clean them. If they are worn out replace with new terminals
- v. Replace the clean terminals and tighten the bolts. Apply petroleum jelly or grease to the connected terminals

Checking and topping up electrolyte

This is done once a month. Check the level of electrolyte in each cell and top up to about 2cm above the plates if necessary. Use de-ionised water or distilled water for topping up. Tap water should not be used.

Check the condition of the plates to see if the active materials are eroded. If eroded the battery needs to be replaced

Checking the state of charge

This should be done once a month. A hydrometer or voltmeter can be used to determine the state of charge. Compare the reading you get to what the manual indicates. If the state of charge is low, allow the battery to be charged fully by the module. If the system is overloaded, reduce the load. Keep record of the state of charge and age of the battery.

Equalizing charge

This should be done once every six months to reduce the electrolyte stratification. This is done by charging the battery well above its normal full charge state using a mains powdered charger. This also assists in removing accumulated sulphates on the plates. It is best done in the cloudiest month of the year.

Module Maintenance

Cleaning the module

Modules do not have any moving parts and will require little maintenance. The main maintenance activity is inspecting for dust and cleaning. This is done monthly or twice a month in dry dusty months. This is done to increase the power output from the module.

The module should be cleaned with water and a soft cloth to avoid scratching it. A scratch can cause shading and reduce the power output of the module.

Tilt angle and orientation should also be checked.

Checking connections of the module

This is done once a year to ensure that the junction box at the back of the module has tightly connected wires. If the wires have been chewed by rodents, they will have to be replaced, spider webs need to be removed.

Wiring and Control Maintenance

The wiring system should be checked at least twice in a year to find out if it has been chewed by rodents, tampered with or if there are loose connections.

The procedure for the maintenance is outlined below.

- i. Check the tightness of all the screws and ensure no bare wires are visible
- ii. Check cable runs for cracks in the insulation. Insulate or replace as necessary
- iii. Check junction boxes to remove insects and to ensure that the boxes are water tight.
- iv. Check switches to ensure they are operating correctly
- v. Check fuses to find out if any has been blown up. If so, find the cause and repair it.
Replace the fuse with one of the same size.
- vi. Check indicator lamps on the control panel to ensure that they are operating correctly
- vii. Check earthing wires to ensure that they are intact

Maintenance of Lamps and other Loads

1. Lamps, reflectors and fixtures should be cleaned regularly to remove dust.
2. Check blackening of fluorescent tubes and replace them.
3. Replace blown lamps

Systems Records and Manuals

The information that should be in these records include the following:

- i. Circuit diagrams and maps that show where batteries, loads, wire runs, junction boxes and buried cables are located
- ii. Manuals, manufacturer's specifications for components and warranties
- iii. Log book with information on battery state of charge and history, installation dates, repairs and equipment replacement and system maintenance schedule.

TROUBLE SHOOTING

Trouble shooting involves identifying problems in the solar PV system and fixing them. The basic equipment needed is a multimeter. The major cause of system failure is usually the battery.

Before embarking on the trouble shooting exercise is important to do basic back ground check. This includes:

- i. How was the weather a week before the problem? If cloudy the battery may not have been charging fully and was overloaded.
- ii. Is the system new and does the owner know how to use it? The system can fail due to some component or improper installation. New owners may also overload the system out of ignorance or excitement.
- iii. What is the condition and age of the battery? If too old \, it could be the cause of the problem
- iv. Check to see if all the fuses and circuit breakers are okay. If they are blown, locate the fault and replace the fuses
- v. Check the wires for corrosion and breaks and for firm connections
- vi. Check the cleanliness of the module

If the problem cannot be fixed with the basic checks then a detailed check can be carried out as shown in the following table

PROBLEM	CAUSE	CORRECTION
Battery state of charge is low (below 11.5V)	No solar charge	Check and fix connection to module
	Battery acid low	Top up with distilled water
	Defective battery or cell	Check state of charge of each cell. If there is significant difference, replace battery
	Loose or corroded battery terminals	Clean and tighten the terminals
	Dusty module	Clean the module
	Over use of system	Recharge the battery elsewhere and leave the appliances OFF for the period of charging
	Battery not charging	Find out age of battery and replace if too old
	High voltage drop between module and battery	Replace cable with larger diameter cable
	Defective charge controller	Replace charge controller
	Short circuit in wires from module	Locate and repair the short circuit

No solar charge	Loose connection between charge controller and battery	Locate and repair the loose connection
	Blown fuse	Locate the fault, correct it and replace fuse
	Dusty module	Clean module
	Broken module	Replace module
Blown fuse	Short circuit along wire to the solar cell module battery ,or load	Locate and repair the short circuit
	Too small a fuse	Use fuse 20% larger than the combined power of the load
	Lightening or power surge	Replace fuse
Appliances and lamps do not work	Lamps	
	Bad tube	Replace tube
	Bad connection of wires	Correct the connection
	Short life span	Check system voltage
	Appliances	Locate broken or loose connection an repair
	Bad socket	Replace socket
	Switch OFF	Turn ON switch
	Broken appliance	Replace/repair appliance

PV SYSTEM SIZING

Definition

Sizing (system design) is a process of getting the correct combination of module size, battery size, charge controller, inverter, cables, and other accessories so as to ensure a properly working system and a satisfied end user. PV systems have to be sized according to the needs of the end user.

Need for sizing a solar PV system

A Solar PV system is sized in order to determine;

- i. The right size of module
- ii. The right size of charge controller
- iii. The right size of battery
- iv. The right size of inverter and
- v. The right size of cables, fuses and circuit breakers.

Proper sizing ensures the best PV-Battery combination to support the end users daily energy design requirement throughout the year while ensuring suitable battery cycling to maximize battery service life.

Oversized systems result in unnecessary additional expense and undersized systems will not regularly deliver the power required leading to batteries being in a consistently low state of charge and consequently shorter battery life and increased operating costs.

Factors to consider when sizing PV systems

- a) End user needs
- b) Losses
- c) Insolation
- d) Acceptable daily depth of discharge (DDOD)
- e) Days of autonomy

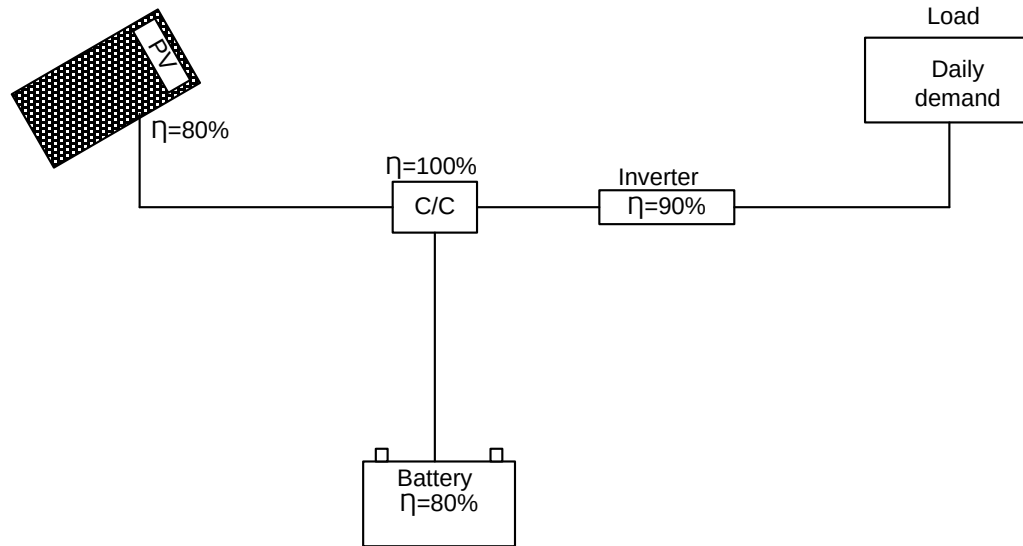
End user needs

It is important that correct information on end user needs is collected outlining required appliances and hours of use.

Losses

Energy is always lost due to inefficiencies in wire, batteries and inverters. This extra amount should be added to the total daily load energy demand. It is difficult to measure energy losses exactly. General efficiency assumptions for various components are as follows:

- i. Conversion of solar energy by PV module is 20%
- ii. PV module output due factors like soiling and internal resistances to be 80%
- iii. Charge controller efficiency to be 100% since it is just a switching device
- iv. Battery efficiency to be 80%
- v. Inverter efficiency to be 90%



Wiring losses are determined by the system voltage, length of cable runs and size of wires. The bigger the wire cross-sectional area, the less the expected losses.

In the solar systems voltage drop is critical due to the DC voltage. Hence use of large cables is recommended and also cable run distances are reduced as much as possible.

Insolation

This refers to the average sun hours falling on a particular location. Insolation data is available from Meteorological department. Insolation maps could be used to determine insolation of a particular location.

5 sun hours per day global insolation is usually acceptable.

Days of autonomy

This refers to the number of days that a fully charged battery/battery bank can supply energy to the system loads when there is no energy supplied by the PV array. For common less critical PV applications, autonomy periods are typically between two and four days.

Acceptable daily depth of discharge (DDOD)

The acceptable DDOD for modified car battery is 50%, for sealed solar batteries is 60% and for flooded deep cycle solar batteries is 80%

PV System sizing /design

This is a process through which one can reasonably determine the size of module(s), battery bank and charge controller required for given daily energy requirements. PV systems are utilized based on the type of electric appliances used by the users. This information is gathered through a site survey. A total energy requirement per day of (watt-hour per day) is estimated by using the calculation sheet.

Steps of the design process

1. Determine the daily energy end user requirements
2. Address system losses and inefficiencies
3. Use appropriate system voltage
4. Size the PV array
5. Determine the battery bank capacity
6. Determine the rating of controller
7. Determine the size of inverter

Determining daily energy demand

Collect end user needs in terms of appliances and daily hours of use. Distinguish which loads/appliances are AC and which are DC and determine the power rating of each of the loads, Then fill the data in the table as shown in Table 10.01.

Appliance	No.	DC or AC	Voltage (V)	Wattage (W)	Daily Hrs. of use	Wattage consumption (Watt-hrs.)
Indoor lights	10	DC	12	9	3	270
Radio	1	DC	12	10	5	50
Colour TV	1	AC	240	40	3	120
Refrigerator	1	AC	240	75	10	750
Total Daily Energy Demand (Wh)						1190 Wh

Table 10.01

Example of solar PV sizing

A homestead in Mtwapa Kilifi County requires an installation of a stand-alone solar PV system to power its appliances. The owner proposes the following load;

Rooms	Appliances	Quantity	Power(W)	Usage hours (h)
Bedroom 1	CFL lamp	1	11	4
Bedroom 2	CFL lamp	1	11	4
Sitting room	LED lamp	2	3	6
	TV	1	75	6
	Decoder	1	30	6
	DVD player	1	30	2
	Phone charger	1	5	2

Kitchen	LED lamp	1	3	4
Security lighting	LED lamp	2	3	10

Additional information

DC-AC inverter efficiency 90%

Battery efficiency 80%

Output performance of PV module 80%

Average insolation in Kenya 5.6kWh/m²/day

Days of autonomy = 3days

Depth of discharge (DOD) = 60%

- a) *Determine the clients daily energy demand*
- b) *Design the system and recommend the ratings and numbers of*
 - i. *PV array*
 - ii. *Batteries*
 - iii. *Charge controller*
 - iv. *Inverter*

Solution

Energy demand estimate

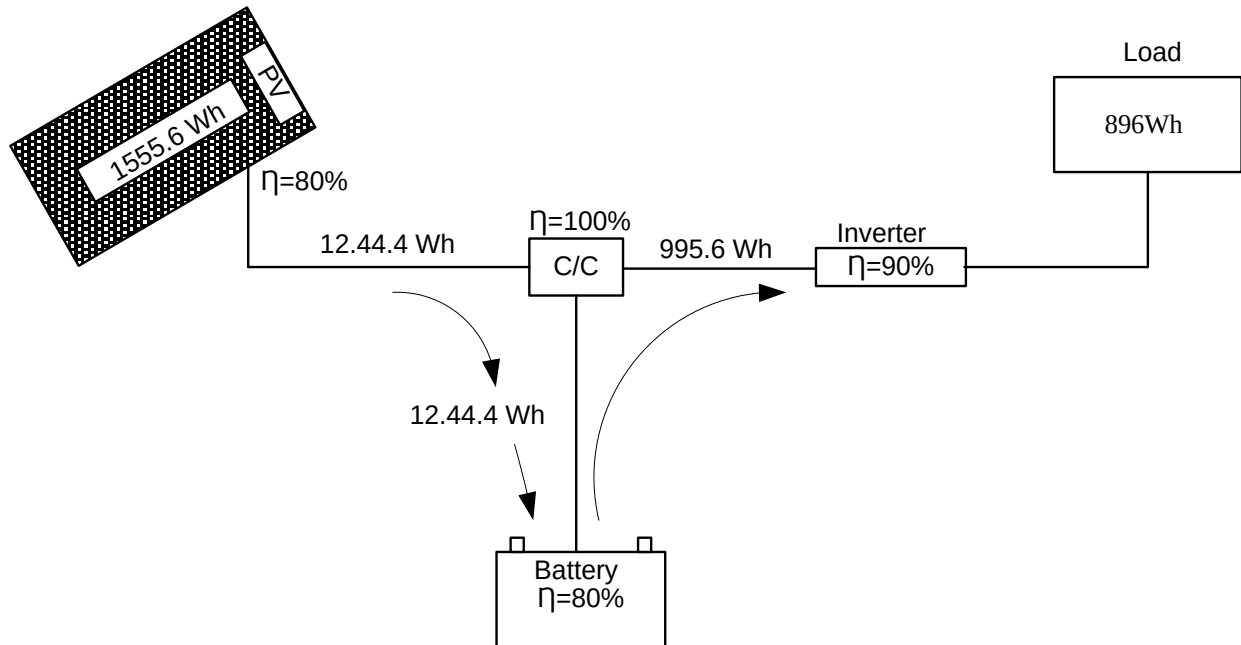
Rooms	Appliances	Quantity	Power(W)	Total Power(W)	Usage hours (h)	Energy consumption (Wh)
Bedroom 1	CFL lamp	1	11	11	4	44
Bedroom 2	CFL lamp	1	11	11	4	44
Sitting room	LED lamp	2	3	6	6	36
	TV	1	75	75	6	450
	Decoder	1	30	30	6	180
	DVD player	1	30	30	2	60
	Phone charger	1	5	5	2	10
Kitchen	LED lamp	1	3	3	4	12
Security lighting	LED lamp	2	3	6	10	60
Peak load(W) =				177	Total energy (Wh) =	896

PV array size

Energy into the inverter = $896 / 0.9 = 995.6 \text{ Wh}$

$$\text{Energy into the battery} = 595.6 / 0.8 = 1244.4 \text{ Wh}$$

$$\text{Energy from the PV} = 1244.4 / 0.8 = 1555.6 \text{ Wh}$$

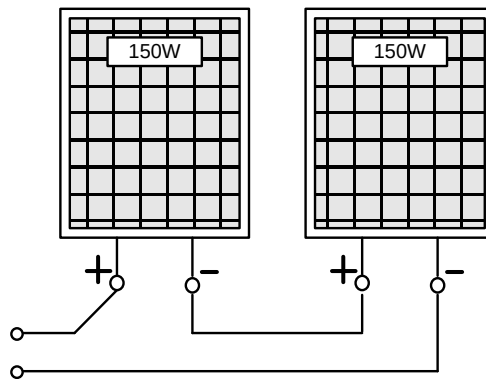


$$\text{PV capacity required} = \frac{1555.6 \text{ Wh}}{\text{insolation}} = \frac{1555.6 \text{ Wh}}{5.6 \text{ h}} = 278 \text{ W}$$

Since the required PV capacity is greater than 150W, then we will use 24V as the system voltage
Therefore, the number of PV modules must be even.

Two PV modules of 150W each totaling 300W are chosen.

The modules will be connected as shown below



Battery capacity

Assumption made

$$\text{DoD} = 60\%$$

The energy into the inverter = the energy leaving the battery =
20% of the battery capacity = 995.6 Wh

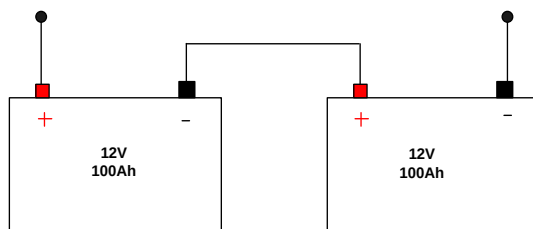
Battery bank capacity therefore = $995.6 \times 100 / 60 = 1659.33\text{Wh}$

Applying days of autonomy $1659.33 \times 3 = 4977.99\text{Wh}$

Dividing this with the bank voltage = $4977.99\text{Wh} / 24\text{V} = 207.41\text{Ah}$

For a 24V system, two batteries of 12V will be connected in series.

In this case 2 batteries of 100Ah each are connected in series as shown below.



Charge controller capacity

To get short circuit current (I_{SC}) at STC , multiply the PV module power rating by 0.064

Therefore,

$$150\text{W} \times 0.064 = 9.6\text{A}$$

I_{SC} for 150W, 9.6A

The modules are connected in series and therefore I_{SC} remains the same

$$\text{Maximum current estimate} = 1.25 \times 9.6 = 12\text{A}$$

Therefore the charge controller should be of 12A and over.

DC-AC inverter capacity

$$= \text{peak load} \times 1.2 = 177\text{W} \times 1.2 = 212.4\text{W}$$

For safety purposes, the inverter should be 25% to 30% bigger

$$212.4\text{W} \times 1.25 = 265.5\text{W}$$

Hence inverter should be 300W and over

ALTERNATIVELY

DC loads are separated from AC loads as below.

Step 1: Determine the Daily Energy End Use Requirement

Load Description	2	3	4	5	6	7	8	9	10	11
Indoor lights	DC	12	10			9	90	3	270	
Radio	DC	12	1			10	10	5	50	
Colour TV	AC	240	1	40	40			3		120
Refrigerator	AC	240	1	75	75			10		750
				Total AC	115	Total DC	100		320	870

				Load		Load				
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Key

1 Load/Appliance description

2 DC or AC

3 Load voltage (V)

4 Quantity

5 AC power rating (W)

6 Total AC power (W)

7 DC power rating (W)

8 Total DC power (W)

9 Daily Use (Hrs/day)

10 Daily DC Energy use (Wh/day)

11 Daily AC Energy use (Wh/day)

Step 2: Address system losses

	Daily Energy Use (Wh)	Power Conversion Efficiency	Battery Efficiency	Array Derating Factor	Wiring Efficiency	Total Daily Energy Use (Wh)
		÷	÷	÷	÷	=
AC Loads	870	0.75	0.9	0.9	0.98	1461.3
DC Loads	320	1	0.9	0.9	0.98	403.1
Corrected Daily Energy Requirement (Wh)						1864.4

Step 3: Choose the Appropriate System Voltage (12 or 24VDC)

The choice of battery voltage depends on the system size. For small applications a 12V system size is suitable. In addition most DC appliances are available in 12V. For the system above a **system voltage of 12VDC** is suitable.

Step 4: Sizing the PV Array

The module/array size is determined based on the daily energy requirement in amp hours and the average daily solar irradiation in the area the system is to be installed. The daily energy

requirement in amp hours is determined by dividing the corrected daily energy requirement by the system voltage.

Corrected Daily Energy Requirement (Watt-hours)	÷ System voltage (VDC)	= Daily Energy requirements (Amps)
1864.4	12	155.37

If the peak sun hour for the selected site is **6.27 peak sun hours**. The module/array current required to provide the daily energy requirement is then determined as shown in the Table below;

Daily Energy Requirement ÷ peak sun hours (Amp hours)	= Required module/array current (Amps)
155.37	24.78

The number of modules connected in parallel to provide this current is determined by the type & model of the module chosen and its rated current e.g. if a 120W Kyocera solar module was selected, 4 modules (3.49 modules rounded up) would have to be connected in parallel to produce the required current. For a 12V system this is the total number of modules required.

Step 5: Determine the Battery Bank Capacity

The recommended days of storage are typically 2-4 days. However caution should be taken to ensure that the array capacity is sufficient to recharge the large battery bank adequately. In cases where the days of autonomy are more than 3 days over sizing the module/array should be considered.

$$\begin{aligned}
 \text{Battery Bank Capacity} &= \text{Daily Energy requirement (Ah)} \times \text{Days of autonomy} \div \text{Maximum DOD} \\
 &= 155.37 \times 2 \div 0.5 (50\%) \\
 &= \mathbf{621.48 Ah/12V}
 \end{aligned}$$

This can be rounded off to the nearest battery size. Remember that batteries of the same capacity, type and model can be connected together in parallel to achieve the required battery capacity.

That is 6 Batteries each rated 100Ah connected in parallel. (6.21 rounded down).

Step 6: Determining the inverter size

$$V_{in} = 12V$$

$$AC \text{ loads} = 75W + 40W = 115W$$

$$\begin{aligned}
 \text{Inverter Rating} &= 115W \times 1.3 = 149.5W \\
 &= \mathbf{150W \text{ or more}}
 \end{aligned}$$

Exercise

A restaurant in a remote part of Kwale which is away from the grid power supply has the following final sub – circuits to be supplied by a solar PV system:

Item	Description	Rating	Number	Voltage level	Usage (hours)
1	Lighting points	10W	10	12V,DC	8
2	Refrigerator	50W	1	240V,AC	12
3	Radio	10W	1	12V,DC	8
4	Colour TV	40W	1	240V,AC	6

The bar and restaurant operates seven days a week. A 5 day system autonomy is required and that a 20% daily depth of discharge is acceptable. An occasional 100% of discharge is acceptable.

Design the solar PV system of the restaurant determining;

- The size of solar array
- The size of battery bank
- The size of charge controller
- The size of inverter